THE INSILICO LAB

3-Week Virtual Training in CADD

Capstone Project

Protein: 7SFB | 5-Ligand Structure-Based Drug Design Pipeline

Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac

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Project Overview

This Capstone Project integrates all skills from Weeks 1–3. Work through the six tasks in order, as each build on the previous. All tables, screenshots, figures, and written justifications are required for full marks.

#	Task	Tool(s)	Key Output
1	Drug-likeness Screening	PubChem + SwissADME	Lipinski table (5 ligands)
2	Protein Preparation	UCSF Chimera / Dockprep	Cleaned 7SFB PDB file
3	Binding Site Prediction	DoGSiteScorer	Binding pocket + residues
4	ADMET Prediction	ADMETlab 3.0	ADMET table (5 ligands)
5	AlphaFold3 Structure Evaluation	AlphaFold3 + PyMOL	Aligned overlay + RMSD

TASK 1 | Drug-likeness Screening

Objective: To evaluate the drug-likeness of all 5 ligands using Lipinski's Rule of Five via SwissADME.

Method Summary:
Drug-likeness screening assesses whether a compound has physicochemical properties consistent with oral bioavailability. Lipinski's Rule of Five (Ro5) states that a drug-like molecule should have: MW ≤ 500 Da, H-bond donors ≤ 5, H-bond acceptors ≤ 10, and LogP ≤ 5. Compounds violating more than one rule are unlikely to be orally bioavailable. SwissADME (swissadme.ch) provides a rapid, web-based implementation of these filters alongside additional ADME predictions.

▸ Task Steps

1	Search for each of the 5 ligands in PubChem (pubchem.ncbi.nlm.nih.gov) and record the CID and canonical SMILES.
2	Go to SwissADME (http://www.swissadme.ch) and paste all 5 SMILES strings (one per line) into the input field.
3	Run the analysis and navigate to the 'Lipinski' and 'Drug-likeness' sections of the results.
4	Record MW, H-bond donors, H-bond acceptors, LogP, rotatable bonds, and overall Ro5 pass/fail for each compound.
5	Download the SDF files from PubChem for each compound.
6	Open and visualize each compound in PyMOL. Take a screenshot of each structure.
7	Take an annotated screenshot of the SwissADME results page showing all 5 compounds.

▸ Deliverable 1A — Compound Information Table

Complete the table below using data from PubChem.

Compound	PubChem CID	Canonical SMILES
Aspirin	**2244**	**CC(=O)OC1=CC=CC=C1C(=O)O**
Ibuprofen	**3672**	**CC(C)CC1=CC=C(C=C1)C(C)C(=O)O**
Acetaminophen	**1983**	**CC(=O)NC1=CC=C(C=C1)O**
Naproxen	**156391**	**C[C@@H](C1=CC2=C(C=C1)C=C(C=C2)OC)C(=O)O**
Diclofenac	**3033**	**C1=CC=C(C(=C1)CC(=O)O)NC2=C(C=CC=C2Cl)Cl**

▸ Deliverable 1B — Lipinski Rule of Five Results

Fill in all values from SwissADME. Use ✓ or ✗ for each rule, then mark overall Pass/Fail.

Compound	MW (≤500)	HBD (≤5)	HBA (≤10)	LogP (≤5)	Ro5 Pass?
Aspirin	**180.16 ✓**	**1 ✓**	**4 ✓**	**1.28 ✓**	**Yes**
Ibuprofen	**206.28 ✓**	**1 ✓**	**2 ✓**	**3.21 ✓**	**Yes**
Acetaminophen	**151.16 ✓**	**2 ✓**	**2 ✓**	**0.86 ✓**	**Yes**
Naproxen	**230.26✓**	**1 ✓**	**3 ✓**	**2.96 ✓**	**Yes**
Diclofenac	**296.15 ✓**	**2 ✓**	**2 ✓**	**4.09 ✓**	**Yes**

▸ Deliverable 1C — PyMOL Structures

Figures

Insert one PyMOL screenshot per compound (5 total). Label each image clearly with the compound name.

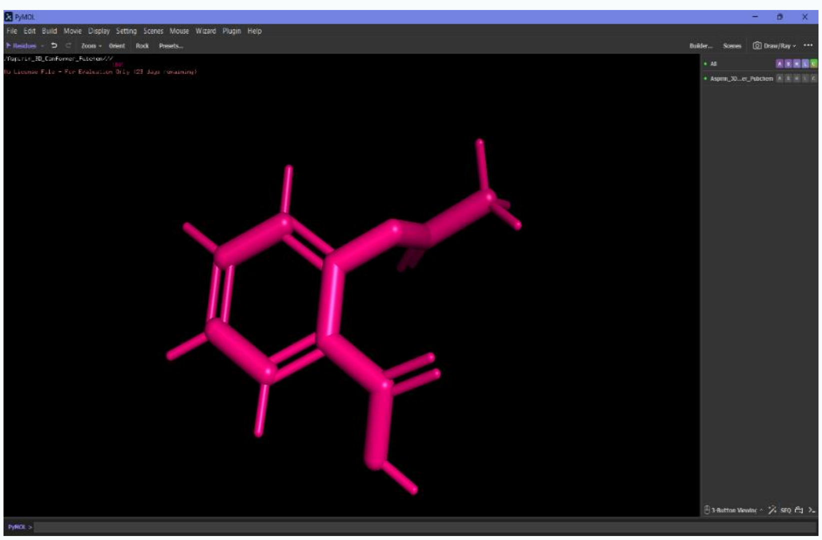


Figure 1.1 — Aspirin Structure (PyMOL)

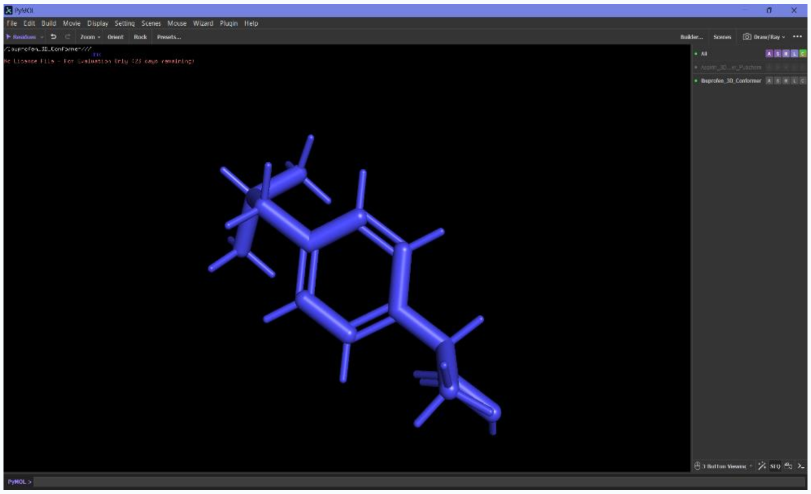


Figure 1.2 — Ibuprofen Structure (PyMOL)

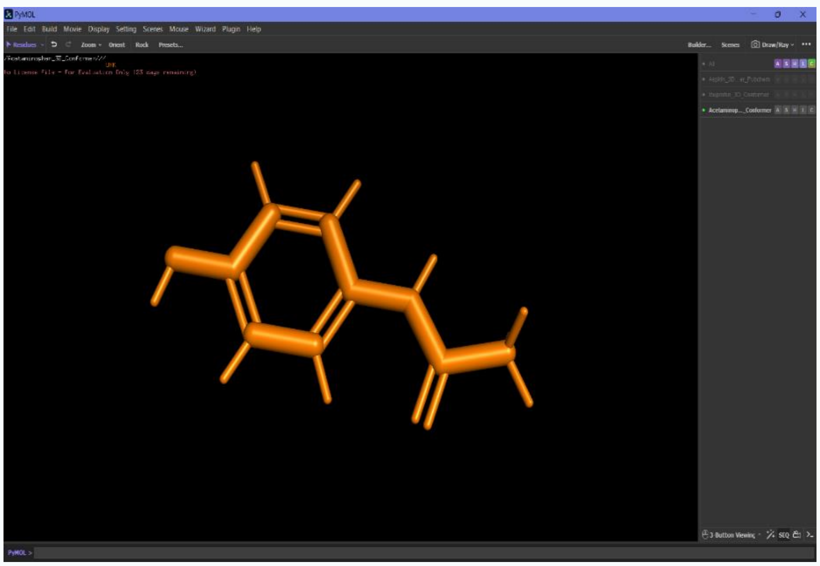


Figure 1.3 — Acetaminophen Structure (PyMOL)

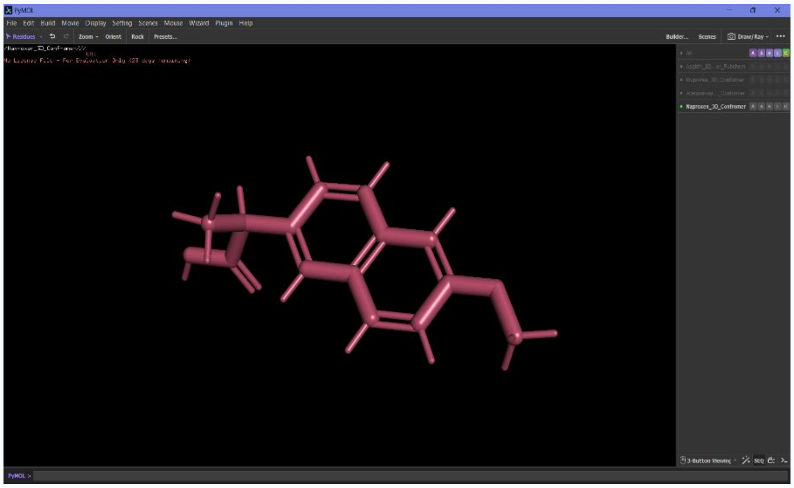


Figure 1.4 — Naproxen Structure (PyMOL)

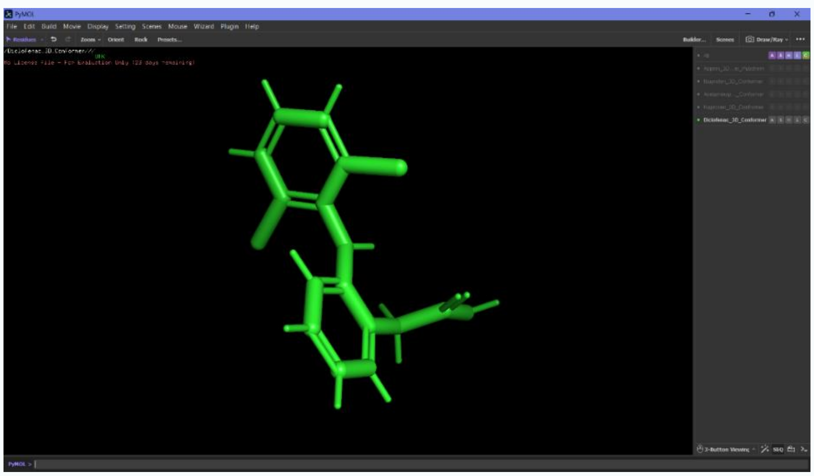


Figure 1.5 — Diclofenac Structure (PyMOL)

▸ Written Justification

Which compounds pass Lipinski's Ro5 and are therefore considered drug-like? Discuss any violations and what they mean for oral bioavailability. (Write 4–6 sentences)

All 5 compounds [Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac] passed Lipinski's Rule of Five and are therefore considered drug-like compounds with good potential for oral bioavailability. None of the compounds violated the key criteria, as all possessed molecular weights below 500 Da, hydrogen bond donors fewer than 5, hydrogen bond acceptors fewer than 10 & LogP values below 5 Since no compound exceeded the permitted thresholds, no Lipinski violations were observed. This Suggests that all 5 ligands are likely to exhibit favorable absorption and permeability when administered orally compounds with relatively higher LogP values, such as Diclofenac and Ibuprofen may show greater lipophilicity.

TASK 2 | Protein Preparation

Objective: To download and prepare the 7SFB protein structure for molecular docking using UCSF Chimera.

Method Summary:
Protein preparation is a critical pre-processing step in structure-based drug design. The raw PDB file downloaded from the Protein Data Bank typically contains water molecules, multiple chains, co-crystallized ligands, and missing atoms that must be addressed before docking. UCSF Chimera's Dock Prep tool automates the addition of hydrogens, assignment of Gasteiger charges, and removal of unwanted molecules, producing a clean, computation-ready structure.

▸ Task Steps

1	Go to RCSB PDB (https://www.rcsb.org) and search for 7SFB. Review the entry details (resolution, method, chains).
2	Download the PDB file for 7SFB (Download Files → PDB Format).
3	Open 7SFB in UCSF Chimera (File → Open).
4	Inspect the structure: note the number of chains, water molecules, ligands, and cofactors present.
5	Select Chain A only (if other chains are present). Delete all other chains.
6	Remove all water molecules (Select → Residue → HOH → Actions → Delete).
7	Remove any unwanted cofactors or heteroatoms not needed for docking.
8	Run Dock Prep (Tools → Structure Editing → Dock Prep). Add hydrogen and Gasteiger charges.
9	Save the prepared structure as 7SFB_prepared.pdb.
10	Open both the raw and prepared files in PyMOL. Take before-and-after screenshots.

Deliverable 2A — Protein Structure Information

Property	Details
PDB ID	**7SFB**
Protein Name	**SARS-CoV-2 Main Protease (Mpro)**
Experimental Method	**X-RAY DIFFRACTION**
Resolution (Å)	**1.90 Å**

Co-crystallized Ligand(s)	**YES**
Water Molecules Removed?	**YES**
Hydrogens Added via Dockprep?	**YES**
Prepared File Name	**7SFB_prepared.pdb**

▸ Deliverable 2B — Step-by-Step Preparation Log

Step	Action Performed	Observation / Result
1	**Searched for 7SFB in PDB**	**Protein identified as SARS-CoV-2 Main Protease (Mpro/3CLpro) with X-ray diffraction method and 1.90 Å resolution**
2	**Downloaded the protein structure in PDB format**	**Raw protein structure file obtained**
3	**Opened 7SFB in UCSF Chimera**	**visualized the Protein structure**
4	**Inspected the structure for chains, ligands, and water molecules**	**Observed protein chains, co-crystallized ligand, and water molecules present in the raw structure**
5	**Selected Chain A**	**Single protein chain was observed**
6	**Removed all water molecules (HOH)**	**Water molecules deleted to avoid interference during docking**
7	**Removed unwanted heteroatoms/cofactors and retained only required protein structure**	**Cleaned protein structure was obtained**
8	**Performed Dock Prep in Chimera**	**Protein was prepared**
9	**Saved the prepared protein as 7SFB_prepared.pdb**	**Prepared docking-ready structure**
10	**Opened raw and prepared structures in PyMOL and captured screenshots**	**Before and after comparison successfully visualized and documented**

▸ Deliverable 2C — Before & After PyMOL Screenshots

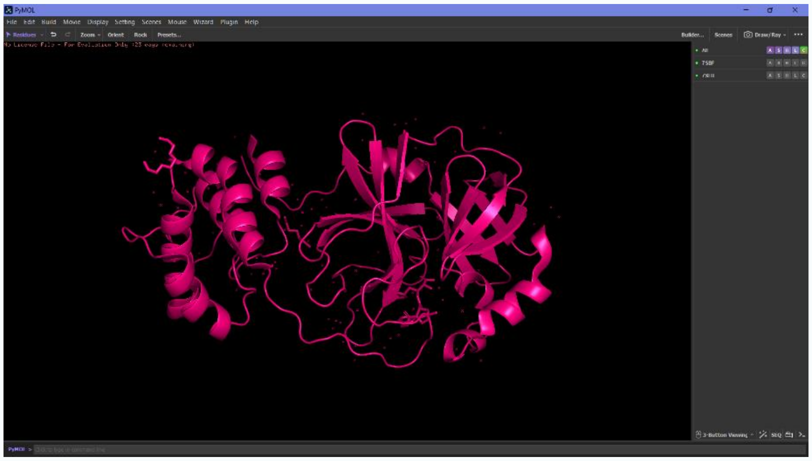


Figure 2.1 — BEFORE Preparation: Raw 7SFB (PyMOL)

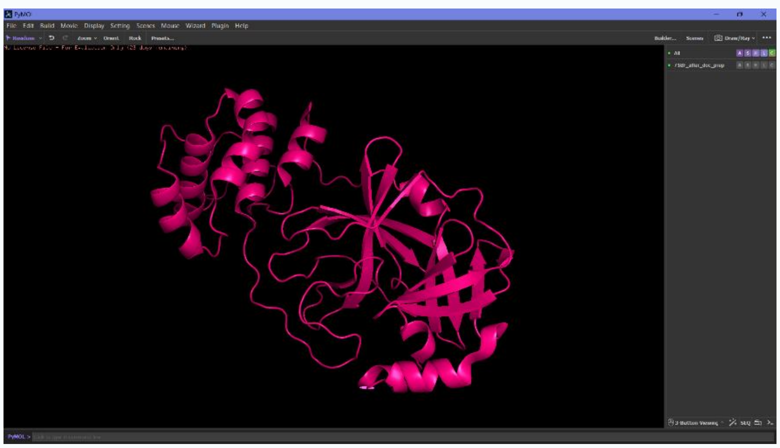


Figure 2.2 — AFTER Preparation: 7SFB_prepared.pdb (PyMOL)

▸ Written Justification

Why is protein preparation necessary before molecular docking? Describe what changes you made to 7SFB and explain the impact of each step. (Write 3–5 sentences)

Protein preparation is necessary before molecular docking to ensure that the protein structure is clean, stable, and suitable or accurate ligand binding predictions. The raw 7SFB structure contained

multiple chains water molecules, and co-crystallized components that could interfere with docking

results so only Chain A was retained while unnecessary chains, water molecules, and unwanted heteroatoms were removed.

Hydrogens were added during Dock Prep to correctly represent protonation states and

hydrogen-bond interactions while charges were assigned to improve electrostatic interaction calculations between the protein and ligands

These preparation steps produced a clean, docking-ready structure (7SFB_prepared.pdb) and increased the reliability and accuracy of molecular docking simulations involving the SARS-CoV-2 Main Protease.

TASK 3 | Binding Site Prediction

Objective: To identify and characterize the primary binding site of 7SFB using DoGSiteScorer on the raw (unprepared) protein.

Method Summary:
Binding site prediction identifies druggable cavities on a protein surface based on geometric and physicochemical properties. DoGSiteScorer (proteins.plus) uses a Difference of Gaussians (DoG) filter to detect pockets and score them by size, hydrophobicity, and druggability. Using the raw (unmodified) protein for this step ensures the prediction reflects the native structure, avoiding artefacts introduced during preparation.

Important
Use the RAW 7SFB PDB file for this task — NOT the prepared version. DoGSiteScorer works best with the unmodified structure.

▸ Task Steps

1	Go to DoGSiteScorer via proteins.plus (https://proteins.plus) and click 'DoGSiteScorer'.
2	Upload the raw 7SFB PDB file (or enter PDB ID directly).
3	Select Chain A (if other chains are present) only in the submission options.
4	Run the prediction and wait for the results.
5	Identify the top-ranked binding pocket (Pocket P_0 or highest-scored).
6	Record the pocket score, volume, surface area, and key residues.
7	Open PyMOL and load the binding pocket. Manually select and highlight the binding site residues.
8	Take PyMOL screenshots showing the pocket surface and labelled residues.

▸ Deliverable 3A — Binding Site Summary

Parameter	Value / Description
Top Pocket ID	**P_0**
Pocket Volume ($Å^3$)	**642.94**
Pocket Surface Area ($Å^2$)	**776.96**
Druggability Score	**0.75**

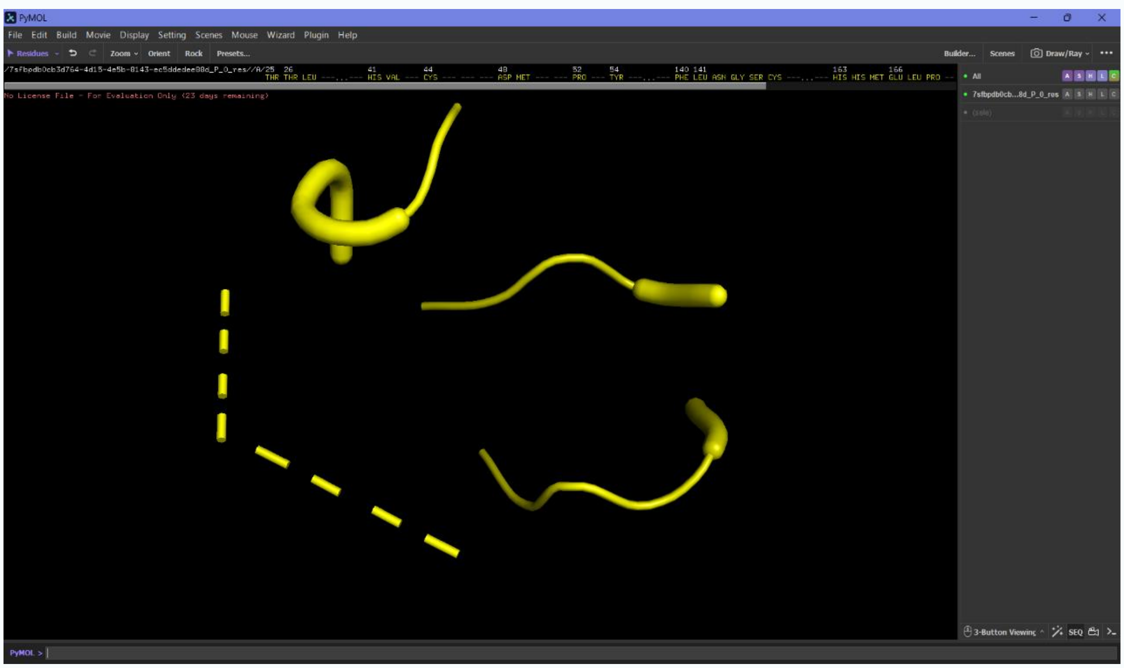
▸ Deliverable 3B — Binding Site Residues Table

List any 5 residues identified in the binding pocket. Include residue name, number

#	Residue Name	Residue Number
1	**THR**	**25**
2	**THR**	**26**
3	**LEU**	**27**
4	**HIS**	**41**
5	**VAL**	**42**

▸ Deliverable 3C — PyMOL Binding Site Visualizations

Figure 3.1 — Binding Pocket Surface View on 7SFB (PyMOL)



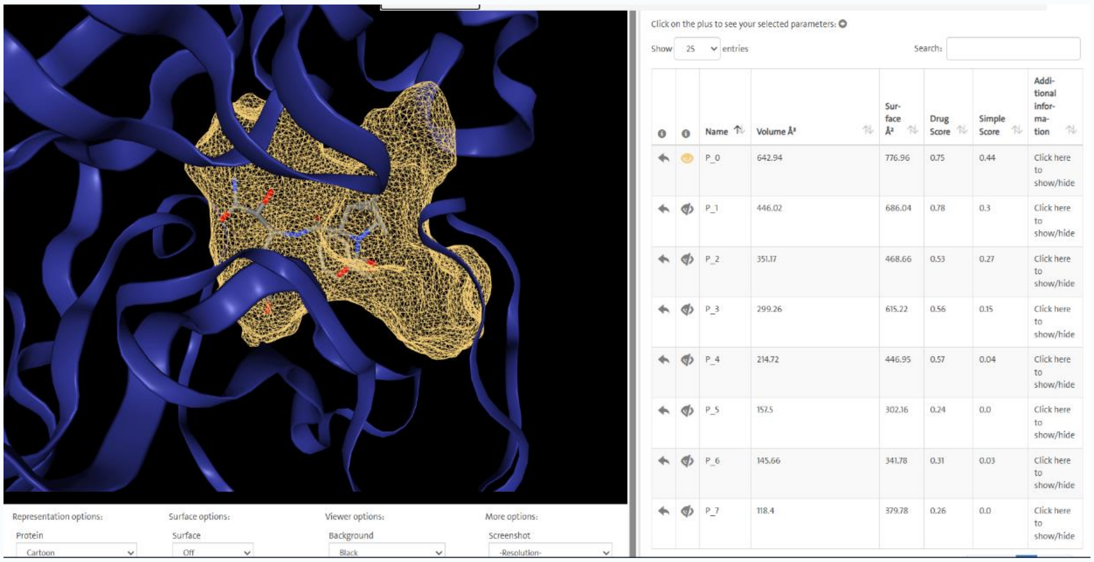


Figure 3.2 — DoGSiteScorer Results Screenshot (proteins.plus website)

▸ Written Justification

Describe the identified binding site. What characteristics make it a good drug target? How will you use these residues to define the docking grid in Task 6? (Write 3–5 sentences)

The identified binding site (Pocket P_0) of the 7SFB showed a pocket volume of 642.94 Å³, surface area

of 776.96 Å² and druggability score of 0.75, indicating a highly favorable region for ligand binding.

The presence of residues such as THR25, THR26, LEU27, HIS41, VAL42, CYS44, and ASP48 suggests

that the pocket contains both polar and hydrophobic amino acids, enabling stable interactions such as

hydrogen bonding and hydrophobic contacts with drug molecules. In particular, HIS41 is important because it is associated with the catalytic region of the protease making it a biologically relevant target for inhibition. In Task 6, these binding site residues will be used to define the docking grid by centering the grid box around the active pocket coordinates to ensure ligands are docked specifically within the predicted binding region for accurate interaction analysis.

TASK 4|ADMET Prediction

Objective: To predict the ADMET profiles of all 5 ligands using ADMETlab 3.0 and identify the most pharmacologically favourable candidates.

Method Summary:
ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) profiling is essential for estimating the in vivo behaviour of drug candidates early in the discovery pipeline. ADMETlab 3.0 (admetlab3.scbdd.com) provides machine-learning-based predictions for over 50 ADMET endpoints from SMILES input. Key properties include Caco-2 permeability, blood–brain barrier penetration, CYP enzyme interactions, half-life, and toxicity flags such as hERG inhibition and AMES mutagenicity.

▸ Task Steps

1	Go to ADMETlab 3.0 (https://admetlab3.scbdd.com) and select 'Prediction'.
2	Paste all 5 SMILES strings (from Task 1) into the input field.
3	Run the full ADMET prediction for all compounds simultaneously.
4	Record at least 3 properties for each of the 5 ADMET categories (15 properties minimum per compound).
5	Take an annotated screenshot of the ADMETlab results page.
6	Download the full prediction report (CSV/Excel if available).

Table Format
The tables below are split into two groups of 5 compounds for readability. Fill in all values from your ADMETlab output.

A Absorption

Compounds 1–5: Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac

A — Property	Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac
Caco-2 Permeability	**-4.985**	**-4.301**	**-4.272**	**-4.657**	**-4.268**
Human Intestinal Absorption (HIA)	**1.66893005371093E-06 (---)**	**0.000122308731079101 (---)**	**0.000263988971710205 (---)**	**0.0000194311141967773 (---)**	**0 (---)**

D Distribution

Compounds 1–5: Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac

D — Property	Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac
Blood-Brain Barrier (BBB)	**0.99768853187561 (+++)**	**0.610769867897033 (+)**	**0.694386780261993 (+)**	**0.00177555298432707 (---)**	**0.997380673885345 (+++)**
Plasma Protein Binding (PPB)	**60.3%**	**97.6%**	**17.3%**	**97.9%**	**99.2%**

M Metabolism

Compounds 1–5: Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac

M — Property	Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac
CYP1A2 Inhibitor	**0.0000488120967929717 (---)**	**2.12991824177777E-09 (---)**	**0.0167834274470806 (---)**	**3.76496586795838E-06 (---)**	**2.13897328649181E-06 (---)**
CYP3A4 Inhibitor	**2.48652412437877E-07 (---)**	**3.23626103693186E-09 (---)**	**0.00345196295529603 (---)**	**6.22237337211117E-08 (---)**	**4.45614212196687E-08 (---)**

E Excretion

Compounds 1–5: Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac

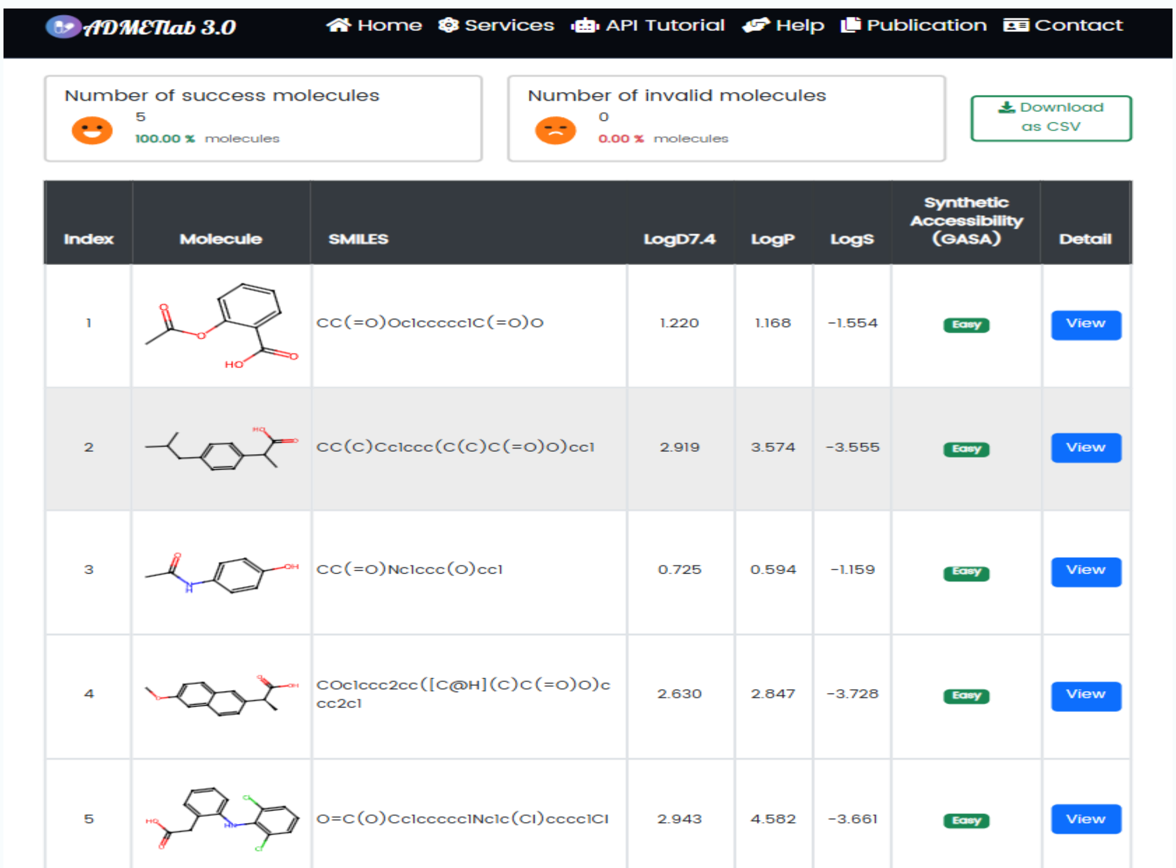
E — Property	Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac
Half-life (T1/2)	**0.822**	**1.449**	**1.115**	**1.786**	**1.64**
Clearance (CL)	**2.766**	**1.038**	**8.903**	**0.375**	**0.458**

Compounds 1–5: Aspirin, Ibuprofen, Acetaminophen, Naproxen, Diclofenac

T — Property	Aspirin	Ibuprofen	Acetaminophen	Naproxen	Diclofenac
hERG Inhibition	**0.015**	**0.165**	**0.07**	**0.167**	**0.069**
Hepatotoxicity (DILI)	**0.744**	**0.928**	**0.619**	**0.998**	**1.0**

▸ Deliverable 4B — ADMETlab Screenshot

Figure 4.1 — ADMETlab 3.0 ADMET Results (All 5 Compounds)



Index	Molecule	SMILES	LogD7.4	LogP	LogS	Synthetic Accessibility (GASA)	Detail
1		CC(=O)Oc1ccccc1C(=O)O	1.220	1.168	-1.554	Easy	View
2		CC(C)Cc1ccc(C(C)C(=O)O)cc1	2.919	3.574	-3.555	Easy	View
3		CC(=O)Nc1ccc(O)cc1	0.725	0.594	-1.159	Easy	View
4		COc1ccc2cc([C@H](C)C(=O)O)ccc2c1	2.630	2.847	-3.728	Easy	View
5		O=C(O)Cc1ccccc1Nc1c(Cl)cccc1Cl	2.943	4.582	-3.661	Easy	View

▸ Written Justification & Comparison

Compare the ADMET profiles across all 5 compounds. Which candidates have the best overall profiles? Highlight critical flags (hERG, hepatotoxicity, BBB) and discuss trade-offs. (Write 5–7 sentences)

Among the five compounds, ibuprofen and aspirin show the most balanced ADMET profiles, with favourable absorption, low CYP inhibition, and manageable toxicity risks. Acetaminophen demonstrates efficient clearance and low plasma protein binding but still carries hepatotoxicity concerns. Naproxen and diclofenac have beneficial pharmacokinetic properties such as longer half-life and strong BBB penetration, respectively, but both show very high hepatotoxicity scores, making them less favourable candidates. Overall, hepatotoxicity (DILI), BBB penetration, and hERG inhibition represent the major trade-offs across the compounds.

TASK 5 | AlphaFold3 AI-Based Structure Evaluation

Objective: To predict the 3D structure of 7SFB using AlphaFold3 and evaluate the prediction quality by aligning it with the cleaned experimental structure from Task 2.

Method Summary:
AlphaFold3 (alphafoldserver.com) is a deep-learning model developed by Google DeepMind that predicts protein 3D structures from amino acid sequences with high accuracy. The predicted structure is coloured by per-residue confidence (pLDDT score): blue = very high confidence (>90), cyan = confident (70–90), yellow = low confidence (50–70), orange = very low (<50). RMSD (Root Mean Square Deviation) quantifies the structural difference between the predicted and experimental models after alignment in PyMOL — values below 2 Å indicate excellent agreement.

▸ Task Steps

1	Go to RCSB PDB and search for 7SFB. Navigate to 'Sequence' and download the FASTA file.
2	Open the FASTA file in a text editor and copy the sequence.
3	Go to AlphaFold3 (https://alphafoldserver.com) and paste the sequence. Submit the prediction job.
4	Once complete, download the predicted structure (CIF or PDB format).
5	Open both the AlphaFold3 predicted structure AND the prepared 7SFB from Task 2 in PyMOL.
6	Align the two structures using the PyMOL command: align predicted_structure, 7SFB_prepared
7	Record the RMSD value reported by PyMOL after alignment.
8	Take a screenshot of the predicted structure alone (coloured by pLDDT).
9	Take a screenshot of the aligned overlay of both structures using contrasting colours.

▸ Deliverable 5A — FASTA Sequence

Paste the FASTA sequence for Chain A of 7SFB in the box below.

```
>7SFB_A   |   Chain A   |   Paste sequence below:
```

>7SFB_1|Chain A|3C-like proteinase|Severe acute respiratory syndrome coronavirus 2 (2697049)

SGFRKMAFPSGKVEGCMVQVTCGTTTLNGLWLDDVVYCPRHVICTSEDMLNPNYEDLLIRKSNHNFLVQAGNVQLRVI

GHSMQNCVLKLKVDTANPKTPKYKFVRIQPGQTFSVLACYNGSPSGVYQCAMRPNFTIKGSFLNGSCGSVGFNIDYDCV

SFCYMHHMELPTGVHAGTDLEGNFYGPFVDRQTAQAAGTDTTITVNVLAWLYAAVINGDRWFLNRFTTTLNDFNLVA

MKYNYEPLTQDHVDILGPLSAQTGIAVLDMCASLKELLQNGMNGRTILGSALLEDEFTPFDVVRQCSGVTFQ

▸ Deliverable 5B — Structural Comparison Summary

Metric	Value	Interpretation
RMSD (Å)	**0.616**	**< 2 Å excellent**
Residues Aligned	**1997 atoms were aligned**	**Higher = better coverage**
Overall Quality	**Excellent**	**Excellent**
Predicted Structure File	**7SFB_alphafold3.cif**	**7SFB_alphafold3.cif**

▸ Deliverable 5C — Screenshots & Figures

Figure 5.1 — FASTA Sequence File of 7SFB (Annotated)

```
>7SFB_1|Chain A|3C-like proteinase|Severe acute respiratory syndrome coronavirus 2 (2697049)
SGFRKMAFPSGKVEGCMVQVTCGTTTLNGLWLDDVVYCPRHVICTSEDMLNPNYEDLLIRKSNHNFLVQAGNVQLRVIGHSMQNCVLKLKVDTANPKTPKYKFVRIQPGQTFSVLACYNGSPSGVYQCAMRPNFTIKGSFLNGSCGSVGFNIDYDCVSFCYMHHMELPTGVHAGTDLEGNFYGPFVDRQT
AQAAGTDTTITVNVLAWLYAAVINGDRWFLNRFTTTLNDFNLVAMKYNYEPLTQDHVDILGPLSAQTGIAVLDMCASLKELLQNGMNGRTILGSALLEDEFTPFDVVRQCSGVTFQ
```

Figure 5.2 — AlphaFold3 Full Website Screenshot (Predicted 7SFB Structure)

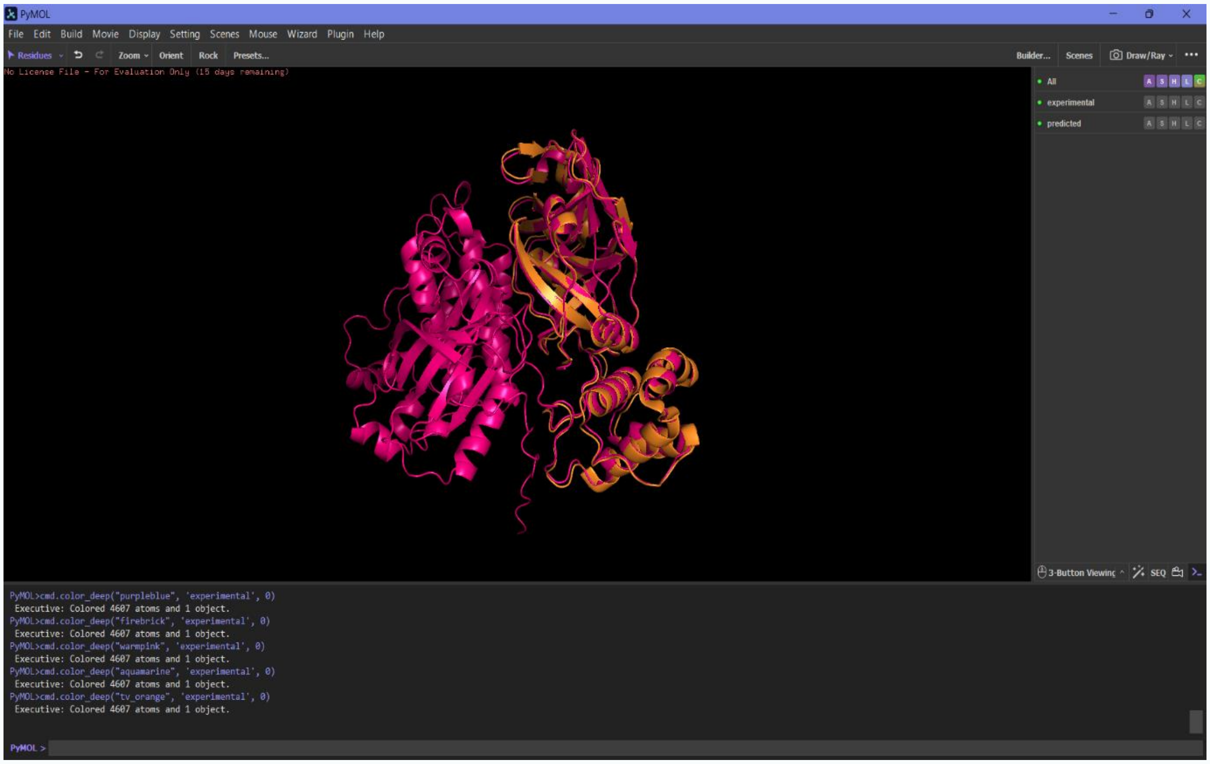


Figure 5.3 — Aligned Overlay: Predicted vs Experimental 7SFB (PyMOL)

▸ Written Justification

How well does the AlphaFold3 structure match the experimental 7SFB structure? Discuss RMSD, confidence scores (pLDDT), and any regions of structural divergence. What does this tell us about AI-based structure prediction? (Write 4–6 sentences)

The AlphaFold3-predicted structure showed excellent agreement with the experimental 7SFB structure, with an RMSD of 0.616 Å, indicating high structural similarity. 1997 atoms were aligned, demonstrating strong overlap between the predicted and experimental models. High pLDDT confidence scores supported the reliability of most structural regions, while minor differences were observed mainly in flexible loop regions. Overall, the results highlight the strong accuracy of AI-based protein structure prediction.

🏆 Capstone Project Complete!

You have successfully completed the full Insilico Lab's CADD pipeline:

Drug-likeness Screening → Protein Preparation → Binding Site Prediction →ADMET →AlphaFold→ Molecular Docking

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End of Capstone Project Template

Good luck! The Insilico Lab Team